

07

Electric Vehicle
Codec Technologies India
Internship 2026

EV Cradle-to-Grave Lifecycle Analysis

Screening-level GHG LCA of a
Nexon-class EV vs petrol SUV
and e-scooter vs petrol scooter,
Python + live-formula Excel

AT A GLANCE

31.5 t CO₂e / 210 g/km

Nexon EV, India grid

41.6 t CO₂e / 277 g/km

Petrol compact SUV

24 / 69 / 79 %

EV reduction India / EU / RE

12,163 km

Car break-even, India grid

2.70 vs 4.33 t (-38 %)

E-scooter vs petrol scooter, India

grid intensity 16.2 t

Largest sensitivity swing

0

Workbook error cells (LibreOffice)

11 / 11

Tests re-run here

VERDICT

TESTS 11/11

RE-RUN HERE

CITED FACTORS

NOT SimaPro / ISO

DECLARED SUBSTITUTION

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1. Objective and brief mapping

Brief: perform a full cradle-to-grave sustainability analysis of an EV using Excel or specialised tools (SimaPro), including materials, energy consumption and recycling impact, and compare with an ICE vehicle. I built a transparent **screening-level attributional GHG LCA (GWP100)** in Python and mirrored every equation as live formulas in an Excel workbook, for two India-relevant pairs: Tata Nexon EV (30.2 kWh, 15.9 kWh/100 km at plug) vs a petrol compact SUV (8.2 L/100 km) over 150,000 km / 12 yr, and a 3.7 kWh e-scooter vs a 110 cc petrol scooter over 60,000 km / 10 yr, under India-grid, EU-grid and 100 % renewable electricity.

Scope statement (README): "It is not an ISO 14040/14044 study, it was not done in SimaPro/GaBi/openLCA, and it uses no ecoinvent inventory. No data in this project is measured by me. Every factor is either cited to a public source (S1-S11) or declared as an assumption." python main.py regenerates results/ in 3 s including a LibreOffice recalculation check of the workbook.

2. Model and architecture

Five phases in kg CO₂e per vehicle: glider = (curb mass - battery mass) x 4.7 (BEV) / 5.2 (ICEV) t CO₂e/t [ICCT 2021 Table A.1]; battery = kWh x 68 kg CO₂e/kWh [ICCT India/China NMC622, range 34-106]; use BEV = km x kWh/km x grid-at-plug with India 727 g/kWh [CEA FY2023-24] x 1.10 upstream / (1 - 0.19 T&D;) = 987 g/kWh, EU 199, RE 35; use ICEV = km x L/km x (0.64 WTT + 2.15 TTW) kg/L; maintenance from ICCT Indonesia; end of life = shredding 30 kg/t + battery recycling process 4 kg/kWh - 20 % credit on battery production. Break-even km = (EV embodied - ICE embodied) / (ICE kg/km - EV kg/km). Battery mass split at 148 Wh/kg.

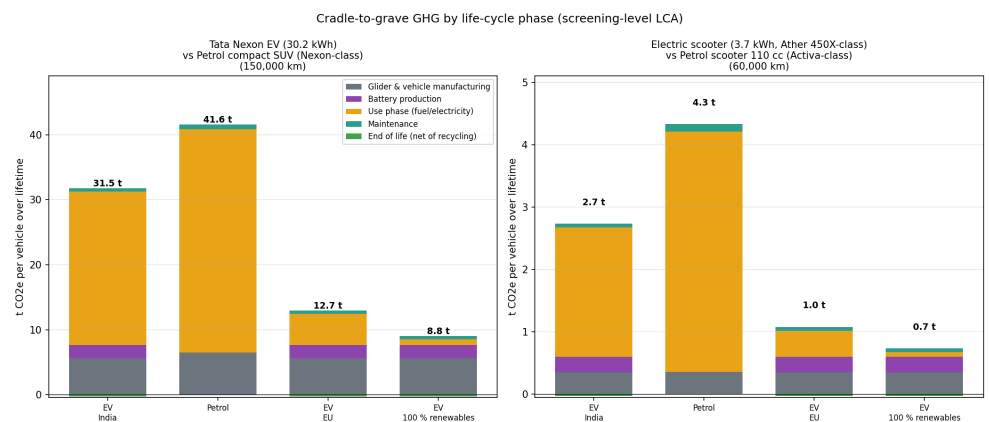


Fig. 1 - Lifecycle GHG by phase for all eight vehicle / scenario combinations (t CO₂e per vehicle over the functional unit).

3. Method and implementation

- src/lca.py computes phases, totals, g/km, break-even and a six-parameter one-at-a-time sensitivity (grid intensity, ICE and EV consumption, lifetime km, battery factor, battery size); src/excel_report.py writes results/ev_lca.xlsx with Inputs, Manufacturing, Use_Phase, End_of_Life, Results, Sensitivity and Sources sheets as formulas.
- Energy: final = kWh x 3.6 MJ or L x 32.2 MJ; primary energy with declared factors (electricity 2.8 India / 2.0 EU / 1.0 RE MJ/MJ, petrol +20 % upstream). Materials: glider split by GREET2-typical shares; battery minerals per kWh scaled from IEA's 75 kWh NMC622 reference; LFP alternative from LiFePO₄ stoichiometry.
- Verification of the workbook: main.py recalculates it headlessly in LibreOffice and scans every cell - 0 error cells; formula totals 31,513 / 41,590 / 2,701 / 4,328 kg equal Python; formula break-even 12,162.8 km equals Python's 12,163.

4. Results (results/lca_results.csv, sensitivity.csv, summary.json)

Break-even distance (car / scooter): India 12,163 / 6,681 km, EU 4,481 / 3,508 km, RE 3,960 / 3,193 km. Sensitivity (car, India, EV-minus-ICE lifetime difference, base -10.1 t): grid intensity 400-900 g/kWh gives -20.7 to -4.5 t (swing 16.2 t); ICE consumption 6.4-9.5 L/100 km -2.5 to -15.5 t (13.0 t); EV consumption 13-19 kWh/100 km -14.4 to -5.5 t (8.9 t); lifetime 100-200k km -6.4 to -13.7 t (7.3 t); battery factor 34-106 kg/kWh -10.9 to -9.2 t (1.7 t); battery size 24-40 kWh 0.4 t. The EV is better than petrol in every one-at-a-time case. Lifetime use-phase energy: Nexon EV 86 GJ final (240 / 172 / 86 GJ primary), petrol SUV 396 GJ final / 475 GJ primary. ICCT's own India result for 2021 cars is 19-38 % lower GHG for BEVs; my 24 % sits inside that range.

Vehicle / scenario	Glider	Battery	Use	Maint.	EoL net	Total t	g/km
Nexon EV - India grid	5.62	2.05	23.55	0.54	-0.25	31.51	210
Nexon EV - EU grid	5.62	2.05	4.75	0.54	-0.25	12.71	85
Nexon EV - 100 % RE	5.62	2.05	0.83	0.54	-0.25	8.80	59
Petrol compact SUV	6.50	0	34.32	0.74	+0.04	41.59	277
E-scooter - India / EU / RE	0.35	0.25	2.07 / 0.42 / 0.07	0.06	-0.03	2.70 / 1.05 / 0.70	45 / 17 / 12
Petrol scooter 110 cc	0.36	0	3.85	0.11	+0.003	4.33	72

KEY TAKEAWAY On today's coal-heavy Indian grid a Nexon-class EV emits 24 % less lifecycle CO₂e than its petrol twin and repays its manufacturing debt in 12,163 km; grid intensity, not battery manufacturing, decides the answer.

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At EU-2027 recovery targets (Li 50 %, Ni/Co/Cu 90 %) the Nexon pack returns about 1.8 kg Li and 19 kg Ni+Co at end of life; India's Battery Waste Management Rules 2022 require 70-80 % overall recovery. Both are computed live in the workbook's End_of_Life sheet.

Material, kg per vehicle	Nexon EV (NMC622)	Nexon EV (LFP alt.)	Petrol SUV	E-scooter	Petrol scooter
Steel / Aluminium / Copper / Plastics	742 / 108 / 36 / 215	742 / 108 / 36 / 215	775 / 113 / 25 / 225	54 / 8 / 3.3 / 16	66 / 10 / 2.1 / 19
Lithium / Nickel / Cobalt / Manganese	3.6 / 16.1 / 5.4 / 9.9	2.6 / 0 / 0 / 0 (+ 20.8 Fe, 11.6 P)	0	0.4 / 2.0 / 0.7 / 1.2	0
Graphite / battery pack mass	27 / 204	27 / 204	0 / 0	3.3 / 25	0 / 0

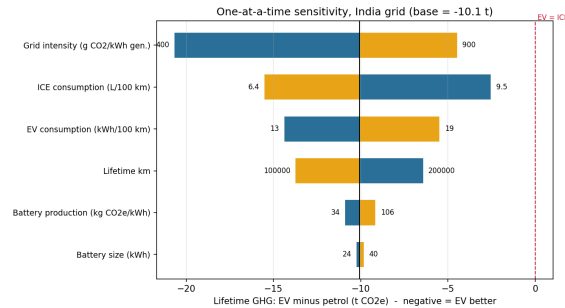


Fig. 2 - Tornado chart, car on the India grid: grid intensity and real-world consumption dominate; battery manufacturing moves the answer by under 2 t.

5. Verification

Re-run here with the fixture-aware runner (pytest not installed): **11 passed, 0 failed** in 3.1 s, including the LibreOffice headless recalculation of the workbook, which is installed in this environment.

Test	Checks	Result
test_phases_sum_to_total, test_summary_json_consistent	five phases sum to totals for all vehicles x scenarios, in code and in summary.json	2 PASS
test_break_even_positive_finite_india, test_break_even_matches_cumulative_curves	break-even positive and finite for car and scooter; the break-even km equalises the cumulative curves	2 PASS
test_ev_below_ice_under_eu_and_india, test_ice_independent_of_grid, test_grid_ordering_and_plug_factor	EV < ICE under EU, India and RE; ICE totals do not depend on grid; RE < EU < India and India plug factor = 727 x 1.10 / 0.81	3 PASS
test_battery_factor_monotonic, test_sensitivity_sorted_and_grid_dominant	battery factor monotone with the exact slope; tornado sorted with grid intensity on top	2 PASS
test_materials_mass_closes	glider material masses close to the glider mass	1 PASS
test_workbook_recalculates_without_errors	LibreOffice recalculation: no #REF!/#DIV/0!/#NAME?/#VALUE!, totals match Python	1 PASS

6. Data integrity and declared substitutions

- **SimaPro / GaBi / openLCA and ecoinvent not used**; this is a screening-level GHG-only study with transparent cited factors, not an ISO 14040/14044 study with a critical review, and covers no acidification, toxicity, water or particulates.
- Proxy factors: glider factors are European (Ricardo / ICCT), maintenance and two-wheeler factors from ICCT's Indonesia study; India-specific manufacturing data was not available to me. The static FY2023-24 CEA grid factor is deliberately conservative for the EV; ICCT's projected India lifetime average (561-746 g/kWh) would improve it by several tonnes.
- My assumptions, each a single cell in Inputs: +10 % upstream uplift for Indian coal, 20 % recycling credit, curb masses 1,400 / 1,250 / 112 / 106 kg, scooter 3.5 kWh/100 km; base battery factor is NMC622 whereas Tata uses LFP (~56 kg/kWh, slightly better for the EV); no battery replacement within 150,000 km.

7. Learnings

Reproducing the numbers: cd 07_EV_Lifecycle_Analysis; python main.py (3 s) rewrites lca_results.csv, sensitivity.csv, materials_kg.csv, summary.json, the four PNGs and ev_lca.xlsx, then recalculates the workbook in LibreOffice headless and compares the formula totals with Python; the same check is test_workbook_recalculates_without_errors, which passed here. Every factor can be changed in src/config.py or in the yellow cells of the Inputs sheet and the whole chain - phases, totals, break-even, tornado - updates.

The factor most people argue about - battery manufacturing - moves the 150,000 km answer by less than 2 t, while grid intensity and the real-world consumption of both vehicles move it by 9-16 t. The EV starts about 0.9 t CO₂e in debt and repays it in the first year of typical Indian driving even on today's grid. Mirroring the model as workbook formulas and recalculating in LibreOffice caught one double-counting risk (the ICCT glider factor already nets glider recycling, so EoL carries only battery and shredding), which I now state in both the code and the workbook.

KEY TAKEAWAY On today's coal-heavy Indian grid a Nexon-class EV emits 24 % less lifecycle CO₂e than its petrol twin and repays its manufacturing debt in 12,163 km; grid intensity, not battery manufacturing, decides the answer.